

Pico Gym Exclusive - Bit-O-Asm-1

Bit-O-Asm-1

Reverse Engineering

Medium

100 pts

by LT 'syreal' Jones

Can you figure out what is in the `eax` register? Put your answer in the picoCTF flag format: `picoCTF{n}` where `n` is the contents of the `eax` register in the decimal number base. If the answer was `0x11` your flag would be `picoCTF{17}`.

In this challenge, we're asked to look at an assembly dump for a program and figure out what's in the `eax` register at the end of the program

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x86 Assembly Instructions

```
<+0>:      endbr64
<+4>:      push    rbp
<+5>:      mov     rbp, rsp
<+8>:      mov     DWORD PTR [rbp-0x4], edi
<+11>:     mov     QWORD PTR [rbp-0x10], rsi
<+15>:     mov     eax, 0x30
<+20>:     pop     rbp
<+21>:     ret
```

To complete this challenge, we'll need to understand what a disassembly dump is

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A disassembly dump is a record of assembly instructions a program executes when it is run

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But to understand the contents of the disassembly dump file, we'll need to understand a bit about the x86 assembly language

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x86 Assembly Instructions

Assembly languages are the programming languages closest to the pure machine code that computer processors execute



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The EAX Memory Register

```
mov     eax, 0x30
```

The goal of the challenge is to understand what value is in the EAX register, but what is it?

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The EAX Memory Register

```
mov     eax, 0x30
```

In x86 assembly, EAX is a memory register, a location where data is stored. The CPU has many memory registers, and EAX is a very commonly-used one

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The MOV Instruction

```
mov     eax, 0x30
```

MOV is an assembly instruction which moves data into memory registers. The syntax is:

MOV <location>, <value>

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The MOV Instruction

```
mov     eax, 0x30
```

So, in this instruction, MOV will move the value 0x30 (hexadecimal 30, which is decimal 48) into the EAX register